

Round Numbers as Attention Windows: Foreign Selling, Retail Counterflow, and Regime-Dependent Volatility in the KOSPI

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June 2026

Draft notice. Working draft for discussion. The corresponding-author role for Y.C.K. is provisional, pending review and consent. Empirical results use a verified KRX investor-type panel; remaining items for final submission are noted in Section 7.

Abstract

Does an equity index behave differently near round-number levels, and if so, who drives it? At the individual-security level, round-number price clustering is a well-documented, largely retail phenomenon (Bloomfield et al., 2025). We ask whether this generalizes to index-level thresholds, where the tradable instruments—ETFs, index futures, program baskets—are dominated by institutional and foreign investors. Using 245 trading days of the KOSPI’s 2025–2026 bull run and a verified Korea Exchange (KRX) panel of daily investor-type net flows, we report three findings. First, absolute returns are modestly larger near round numbers under a relative-distance definition (Mann–Whitney $p = 0.027$), but this vanishes

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under the standard absolute M -value definition ($p = 0.58$) and over long horizons—the effect is regime-dependent and definition-sensitive. Second, foreign net selling strongly predicts next-day index declines (coef. -0.0057 , $p = 0.002$, HAC), while round-number proximity itself does not ($p = 0.26$), and the round $\times$ foreign-selling interaction is insignificant ($p = 0.30$). Third, and decisively, a matched-control design—pairing round-zone days with non-round days of similar trend, volatility, momentum, and drawdown—finds *no* round-specific actor effect: foreign investors actually sold somewhat *more* in matched non-round windows. We therefore reframe index round numbers not as a distinct cause of selling but as *attention windows* in which foreign supply shocks and market overheating become salient. This is a more cautious claim than a “scale paradox,” and we argue it is the defensible one.

Keywords: round numbers; attention; investor flows; foreign selling; regime dependence; KOSPI

JEL: G12; G14; G41; C58

1 Introduction

Round numbers coordinate attention, order placement, and algorithmic execution. At the level of an individual security, the clustering of trades at round prices is robustly documented and is primarily a retail-investor phenomenon driven by cognitive salience (Harris, 1991; Bloomfield et al., 2025). Whether the same logic extends to a *broad index*—KOSPI 5,000 or 8,000—is far less clear. An index is not submitted as a limit order by retail traders; index-level pressure is transmitted through futures, ETFs, index arbitrage, and program trading, instruments dominated by institutional and foreign participants. This gap motivates our question: when the reference point shifts from a security price to an index level, does the round-number effect persist, and who drives it?

The 2025–2026 KOSPI bull run, in which the index rose roughly 2.8-fold within a year and crossed six 1,000-point levels, provides a natural setting. Our prior expectation, which we state and then revise, was a “scale inversion”: that foreign and institutional investors sell at index

round numbers while retail investors absorb. The data partially motivate this picture but do not support its strong, causal form. Instead, the evidence points to a more measured conclusion: round numbers are *attention windows* that coincide with, but do not independently cause, foreign supply shocks during an overheated regime.

2 Data and Methods

2.1 Data

We use daily KOSPI closing levels (FinanceDataReader) and a verified KRX panel of daily net trading values for individual, foreign, and institutional investors, spanning 4 June 2025 to 5 June 2026 ($n = 245$). The round-zone share ($|d_t| \leq 3\%$) is 32.7%. We adopt the sign convention that net selling equals the negative of net buying, in KRW trillion.

2.2 Variables and tests

For each day we compute the log return r_t , the relative distance to the nearest 1,000-point level d_t , a round-pressure intensity $RP_t = \exp[-\frac{1}{2}(d_t/0.03)^2]$, the M -value (absolute trailing digit), trailing 5- and 20-day volatility, and 20-day momentum. We run: (i) volatility comparisons (Mann–Whitney, Levene) under both relative-distance and M -value definitions; (ii) the Aggarwal–Lucey barrier battery; (iii) a next-day return regression on round pressure, controls, and investor net selling with HAC errors; (iv) an interaction model (round $\times$ actor selling); and (v) a matched-control design pairing round-zone days with non-round days of similar trend, volatility, momentum, turnover, and drawdown. Because the three investor flows are near-collinear (they approximately sum to zero), we report a baseline with individuals as the omitted category and flag variance-inflation diagnostics rather than interpreting a full three-actor model.

3 Results

3.1 A definition-sensitive, regime-dependent volatility effect

Near-round absolute returns exceed far-round ones under the relative-distance definition (volatility ratio 1.25, Mann–Whitney $p = 0.027$), and the effect is robust to round-band width ($\pm 2/3/5\%$) and survives a placebo test (a pseudo-barrier shifted by 250 points is insignificant). However, it disappears under the standard M -value definition (ratio 0.57, $p = 0.58$) and over a ten-year sample ($p = 0.89$). The divergence is mechanical: during a 2.8-fold ascent a fixed percentage band maps to a moving point band, so the two definitions select different observations. We treat this definition sensitivity as a methodological result in its own right.

3.2 Foreign selling, not round proximity, predicts declines

Table 1 reports the next-day return regression. Foreign net selling significantly predicts lower next-day returns (coef. -0.0057 , $p = 0.002$), while round pressure itself is insignificant ($p = 0.26$) and institutional selling is not significant on its own ($p = 0.28$). Short-run reversal (r_t , $p = 0.01$) and 20-day momentum ($p = 0.02$) enter as expected. In the interaction model, the round $\times$ foreign-selling term is insignificant ($p = 0.30$): foreign selling matters, but not *more so* near round numbers.

Table 1: Next-day return regression (HAC errors, $n = 185$).

Variable	Coef.	p	Note
Round pressure	+0.0069	0.26	n.s.
Foreign net selling (z)	-0.0057	0.002	predicts decline
Institutional net selling (z)	-0.0016	0.28	n.s.
r_t (1-day)	-0.272	0.01	reversal
r_t (20-day)	+0.113	0.02	momentum

3.3 Matched control: no round-specific actor effect

Table 2 is the decisive test. Matching round-zone days to non-round days on trend, volatility, momentum, and drawdown (balance achieved; all covariate differences insignificant), we find

no significant round-specific selling for any actor. Strikingly, foreign investors net sold *more* in matched non-round windows (0.89 vs. 0.57 KRW tn; difference $p = 0.31$). The round-zone selling pattern therefore appears to reflect high-level profit-taking under overheated conditions rather than a distinct round-number mechanism.

Table 2: Matched-control actor effects (net selling, KRW tn).

Actor	Round zone	Matched control	Difference	p
Individual	-0.118	-0.540	+0.422	0.31
Foreign	+0.569	+0.894	-0.325	0.31
Institution	-0.405	-0.260	-0.144	0.50

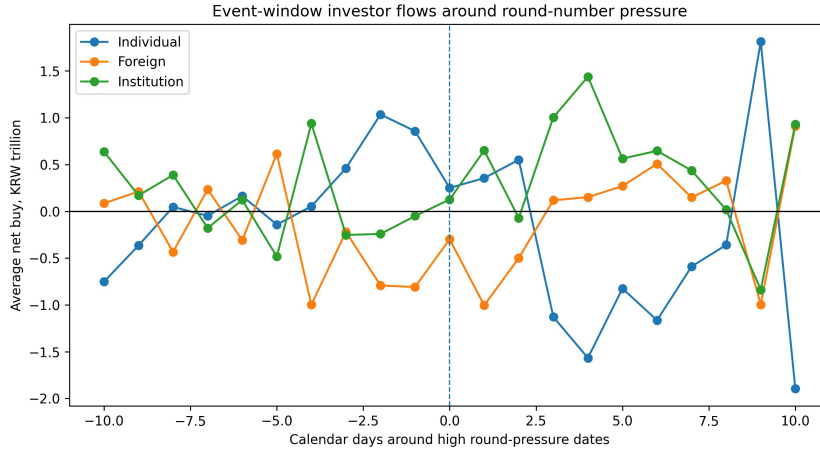


Figure 1: Event-window net flows (± 10 days) around high round-pressure dates. Flows are noisy and show no clean actor switch at the threshold (day 0), consistent with the matched-control null.

4 Discussion

Our initial conjecture—a scale inversion in which foreign and institutional investors drive round-number selling—is only weakly and non-significantly supported in direction, and is rejected once market conditions are matched. What survives is sharper and more credible: foreign net selling is a strong predictor of next-day index declines, but its force is not amplified by round-number proximity. We therefore interpret index round numbers as *attention windows*: salient coordinates at which overheating and foreign supply shocks become observable, rather than independent causes of selling. This reframing aligns with Korean evidence that foreign and

institutional herding is largely information-driven (Kim et al., 2020), and it explains why long-horizon studies of index “barriers” so often return nulls.

5 Limitations and Next Steps

The analysis covers a single market and a single bull-run regime. Three extensions would strengthen it: (i) intraday futures and program-trading data to test the derivative channel directly and to address causal direction; (ii) cross-market replication (other Asia-Pacific indices, prior bull runs); and (iii) an options-gamma analysis at round strikes. The investor panel is daily; intraday flows may reveal threshold dynamics averaged out at daily frequency.

6 Conclusion

Index round numbers in the KOSPI are best understood not as causes of selling but as attention windows in which foreign supply shocks and market overheating become salient. Foreign net selling robustly predicts next-day declines; round-number proximity does not, and adds no amplification once conditions are matched. The contribution is a disciplined negative result paired with a robust predictor—an honest reframing that we believe travels further than the stronger claim it replaces.

Data and Code Availability

The KOSPI series, KRX investor panel, and all Python scripts reproducing the tables and figures are available from the authors on request.

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